OpenReview.net(/)

← Go to **ICLR 2027 Conference** homepage (/group?id=ICLR.cc/2027/Conference)

2 Versions

AttriSelect: Efficient Attribute-Driven Data Selection for Targeted Vision Model Repair

(/pdf?id=n5SiiHVJb8)

Ouyang Xu (/profile?id=~Ouyang_Xu1)[1], Baoming Zhang (/profile?id=~Baoming_Zhang2)[1], RUIYU MAO (/profile?id=~RUIYU_MA[1], Yunhui Guo (/profile?id=~Yunhui_Guo2)[1]

[1] University of Texas at Dallas (utdallas.edu)

20 Aug 2026 (modified: 15 Sept 2026) ICLR 2027 Conference Submission Conference, Ouyang Xu, Baoming Zhang, RUIYU MAO, Yunhui Guo
Edit

Revisions (/revisions?id=n5SiiHVJb8) CC BY 4.0 (https://creativecommons.org/licenses/by/4.0/)

Keywords: vision model repair, data subset selection, active learning

Abstract:
Targeted vision repair aims to correct model failures on rare semantic slices after a debugging pipeline identifies the failure slice and a generator produces corresponding repair images. In most exist workflows, these repair images are appended to the original training set, and the model is retrained or fine-tuned using all source data. While this preserves broad coverage, it also makes each repai computationally expensive and time-consuming, especially as repairs accumulate over multiple iterations. Moreover, it provides little control over which source examples actually contribute to the rep poses a data selection problem related to active learning: rather than using the full dataset, the goal is to identify a compact yet informative subset that best supports repair while substantially reduc retraining cost. However, an effective repair subset must satisfy requirements beyond standard sample efficiency. In particular, it should preserve class coverage, retain rare repair-relevant attributes, avoid redundancy with the generated repair data. Existing active learning methods typically prioritize uncertainty, diversity, or gradient-space coverage, but these signals are often poorly aligned with semantic attributes that define the failure slice. To address this limitation, we propose AttriSelect, an attribute-driven selector that prioritizes samples based on attribute rarity, generated-data covera soft class balance. We provide theoretical analysis and evaluate AttriSelect across image classification, pose estimation, and object detection. Across tasks and repair budgets, AttriSelect gives the stro repair results in most settings, showing that explicit attribute metadata provides a repair-specific selection signal distinct from traditional active learning criteria.

Primary Area: applications to computer vision, audio, language, and other modalities
Code Of Ethics: I and all co-authors of this work have read, and commit to adhering to the ICLR Code of Ethics.
Paper Visibility: I and all co-authors of this work understand that all papers submitted to ICLR become public at the beginning of the review process, and both accepted and rejected papers will be c anonymized at the end of the review process. (Withdrawn papers are de-anonymized immediately upon withdrawal.) We acknowledge that papers cannot be deleted, hidden, or retracted once review begins.
Submission Requirements: I and all co-authors agree to abide by the reciprocal reviewing requirement, the AI disclosure requirement, and all other submission instructions described in https://iclr.cc/Conferences/2027/AuthorGuidelines (https://iclr.cc/Conferences/2027/AuthorGuidelines).
Reciprocal Reviewing Author: Ouyang Xu (/profile?id=~Ouyang_Xu1)
Reciprocal Reviewing Exemption: We do not need an exemption.
AI Assistance: Yes, to aid or polish writing. Details are described in the paper., Yes, for research ideation or execution. Details are described in the paper., Yes, for generating synthetic datasets. De described in the paper., Yes, for proving mathematical claims. Details are described in the paper.
Ready For LLM Feedback: The submitted PDF is ready for LLM feedback.
LLM Feedback Request: Ouyang Xu (/profile?id=~Ouyang_Xu1)
Submission Number: 1973

Filter by reply type... Filter by author... Search keywords... Sort: Newest First

Everyone Conference Submission1973 Authors

2 / 2 replies

LLM Feedback by Program Chairs

LLM Feedback by Program Chairs 15 Sept 2026, 23:14 Conference, Authors

Feedback:
PATLibraryRunPipeline:

HIGH LEVEL SUMMARY

Summary

This paper investigates targeted vision model repair, focusing on the computational bottleneck encountered during the retraining phase. In conventional generative debugging workflows (such as SafeFix and HiBug), targeted synthetic repair examples are generated for an identified failure slice and appended to the original training pool, requiring full-dataset retraining or fine-tuning to prevent catastrophic forgetting. The authors reformulate this retraining stage as a repair-time source-data selection problem: selecting a compact, representative subset of the original source dataset to accompany the synthetic repair instances. To address this challenge, the authors propose AttriSelect, an attribute-driven selection algorithm that scores candidate source examples by combining three complementary terms: (1) class-conditional marginal attribute rarity to retain rare, repair-relevant real examples; (2) a generated-data frequency penalty to suppress redundancy with synthetic images; and (3) a dynamic soft class-balancing bonus to maintain broad semantic coverage across nominal classes. The authors provide theoretical analysis characterizing the base-rate limitations of slice-agnostic active learning methods (Proposition 1) and proving deterministic slice enrichment under bounded penalty conditions (Propositions 2 and 3). The method is empirically evaluated across image classification (ImageNet10, CelebA), multi-person pose estimation (COCO), and object detection (KITTI), demonstrating that compact source subsets (down to 10%–50% of the original pool) can match or outperform full-dataset retraining while executing online subset selection in under two seconds.

Strengths

- The paper introduces a compelling and practically grounded problem formulation by casting targeted vision model repair as a repair-time source-data subset selection task. Decoupling the downstream selection problem from slice discovery and image generation directly targets the core retraining bottleneck in iterative model debugging workflows.
- The proposed scoring objective is principled, interpretable, and computationally elegant. Combining class-conditional attribute rarity, a synthetic-pool redundancy penalty, and dynamic soft cla balancing effectively addresses attribute saturation and class starvation without requiring rigid integer quota constraints.
- The framework is extensively evaluated across diverse computer vision modalities, spanning multi-class image classification (ImageNet10), binary facial attribute classification (CelebA), dense keypoint estimation (COCO), and 2D bounding-box detection (KITTI), demonstrating consistent improvements over generic active learning heuristics across multiple budget ratios.
- The theoretical analysis provides valuable formal insights, notably formalizing the base-rate mismatch inherent in slice-agnostic active learning selectors via conditional independence (Proposition 1) and establishing deterministic target-slice enrichment bounds under the rarity-dominates-penalty condition (Propositions 2 and 3).

- The online selection mechanism achieves remarkable computational efficiency, selecting source subsets in under two seconds on multi-thousand image pools by operating directly over discrete categorical counts and pre-sorted queues, avoiding expensive forward/backward passes or iterative continuous kernel evaluations.
- The authors conduct extensive supplementary experiments, including ablations across score components, sensitivity sweeps over hyperparameters and training schedules, cross-generator evaluations (InstructPix2Pix, ICEdit), human audits of VLM annotations, and evaluations in standalone active learning settings.

Weaknesses

- The experimental evaluation compares AttriSelect primarily against generic, slice-agnostic active learning baselines (such as CoreSet, BADGE, Entropy, PPAL, and Uncertainty Herding). While Proposition 1 demonstrates that slice-blind methods naturally suffer from base-rate limits, the empirical validation would be substantially more persuasive if the authors benchmarked against established targeted active learning and submodular mutual information methods specifically designed to mine rare slices and handle distribution shifts, such as TALISMAN and ORIENT (Reference Paper: TALISMAN: Targeted Active Learning for Object Detection with Rare Classes and Slices Using Submodular Mutual Information, 2022) (Reference Paper: ORIENT: Submodular Mutual Information Measures for Data Subset Selection under Distribution Shift, 2022).
- The paper does not compare AttriSelect against validation-targeted coreset selection and gradient-matching frameworks that optimize training subsets directly to match target validation gradients or maximize performance on target slices, such as GRAD-MATCH or GLISTER (Reference Paper: GRAD-MATCH: Gradient Matching based Data Subset Selection for Efficient Deep Model Training, 2021) (Reference Paper: GLISTER: Generalization Based Data Subset Selection for Efficient and Robust Learning, 2021). Including or discussing these targeted subset selection paradigms would clarify where discrete count-based scoring stands relative to continuous gradient optimization.
- The empirical benchmarks rely on custom task setups that differ from canonical subpopulation and autonomous driving evaluations. Specifically, CelebA is evaluated on an ad-hoc 'lipstick-wearing' task rather than the canonical worst-group blond hair benchmark studied extensively in the subpopulation shift literature (Reference Paper: Distributionally Robust Neural Networks for Group Shifts: On the Importance of Regularization for Worst-Case Generalization, 2020), and object detection is evaluated solely on the small-scale KITTI dataset (5,000 images, where severe low-data collapse occurs at ratio 0.1) rather than established large-scale driving benchmarks with rare condition slices such as BDD100K (Reference Paper: BDD100K: A Diverse Driving Dataset for Heterogeneous Multitask Learning, 2020). Evaluating on these standard benchmarks would help isolate method performance on recognized subpopulation shifts.
- The operational definitions of class labels (y_i), class counts ($n_c^{(k)}$), and soft class balancing ($C_c^{(k)}$) are ambiguous for dense prediction tasks. In COCO person keypoint estimation, all candidate images belong to a single category ('person', implying $N_{\text{cls}} = 1$), which mathematically causes the additive class bonus $C_1^{(k)}$ to be identical across all remaining candidates at step k, having no effect on candidate rankings; yet, Table 33 sets $w = 15$ and Table 36 demonstrates that removing C noticeably alters AP. Similarly, for KITTI object detection, road scenes contain co-occurring instances of both Cars and Pedestrians. Please consider clarifying how candidate images are mapped to classes or pseudo-groups in dense tasks and how multi-instance class quotas are tracked.
- The primary headline results in Table 1 report Relative Error Reduction (RER) computed exclusively over the entire unsegmented test set, where the diagnosed failure slice represents only a small fraction of evaluation images (~1.2% in ImageNet10, ~1.8% in COCO, and two images per seed in KITTI). Furthermore, Table 17 reveals that on the exact ImageNet10 failure conjunction, simpler heuristic baselines achieve higher raw accuracy than AttriSelect, while AttriSelect achieves higher overall RER due to higher accuracy on the non-target population. The authors might consider reporting target-slice metrics (such as slice RER or slice AP) alongside overall test RER across all budget ratios to provide a clear view of failure repair versus nominal regularization.
- The selection policy exhibits vulnerability to attribute corruption at higher noise levels. In Table 21, introducing 20% attribute noise causes an absolute performance drop of 9.0 percentage points on CelebA and 6.0 mAP points on KITTI, dropping performance below the un-repaired checkpoint. Because the class-conditional marginal rarity term assigns maximum weight to low-frequency attribute tokens, corrupted or hallucinated tokens naturally generate artificial singletons that the selector aggressively prioritizes. The authors are encouraged to discuss this noise-magnification characteristic in the limitations and explore potential mitigations, such as continuous VLM confidence weighting or minimum frequency thresholds.
- While Table 2 highlights sub-second online subset selection speeds, the method requires an offline preprocessing phase to extract structured attribute records using Qwen3-VL, which takes approximately 30 minutes for ImageNet10 and KITTI, and 90 minutes for COCO on an NVIDIA H200 GPU. In contrast, standard active learning baselines do not require auxiliary VLM annotation. While this cost is documented in Appendix B.1, presenting the upfront VLM tagging overhead directly in Section 5.2 alongside Table 2 and discussing the operational break-even point across repeated repair cycles would provide a more complete and transparent systems perspective.
- The experimental evaluation is restricted to static, single-round repair interventions. Given that the paper explicitly motivates the method by the compounding computational cost as repairs accumulate across successive debugging cycles, it remains unverified whether repeatedly discarding 50%–90% of source data across sequential repair iterations induces representation drift or catastrophic forgetting on previously stabilized nominal slices. Evaluating a multi-cycle repair benchmark would substantially reinforce the paper's core motivation.

Potential Issues And Suggestions

[Problem Formulation, Failure Slices, and Related Work] (Pages: 1-3)

Summary

This segment (comprising the Abstract, Section 1: Introduction, Section 2: Related Work, Section 3: Background, and the opening of Section 4.1) formulates the problem of targeted vision model repair as a repair-time source-data selection task. Rather than retraining or fine-tuning a model on the full union of the original training set (D_{src}) and newly synthesized repair examples (D_{gen})—which is computationally demanding for dense perception architectures—the authors propose selecting a compact source subset ($D_{\text{sel}}^{\text{src}} \subseteq D_{\text{src}}$) to accompany D_{gen} during retraining.

The segment positions this task at the intersection of two areas: (1) attribute-based model debugging and generative repair pipelines (e.g., HiBug, SafeFix), which focus on diagnosing failure slices and generating targeted synthetic data, and (2) active learning (AL) and data subset selection methods (e.g., CoreSet, BADGE, PPAL, Uncertainty Herding). The text argues that standard AL heuristics are subject to a base-rate mismatch on rare slices (formalized in Proposition 1) and fail to account for redundancy against synthetic repair data. The authors propose AttriSelect, an attribute-driven selector that scores source candidates by combining class-conditional marginal rarity, a generated-data frequency penalty, and soft class balancing.

Potential Mistakes and Improvements

1. **Characterization of Retraining Cost in Problem Motivation**
   - *Context:* In Section 1 (lines 045–049), the paper motivates source-data selection by stating: *"Retraining on the full source pool can be costly, especially for dense prediction tasks. For example, training Keypoint R-CNN for COCO person keypoint estimation (He et al., 2017; Lin et al., 2014) requires approximately eight GPU-days (Facebook AI Research, 2024)."*
   - *Potential Concern:* The eight GPU-day figure corresponds to training Keypoint R-CNN from scratch. However, the repair pipeline described in Section 4.1 and Appendix B (lines 748–751) fine-tunes an existing checkpoint θ_t for a reduced schedule (e.g., 15 epochs at a learning rate of 1×10^{-5}, half of the diagnosis training schedule). As shown in Table 8, full-source fine-tuning takes between 1.5 and 3 GPU-hours rather than multiple GPU-days. Citing the cost of from-scratch training overstates the baseline repair retraining bottleneck.
   - *Suggested Improvement:* Adjust lines 045–049 to state the actual GPU-hour or wall-clock cost of full-source fine-tuning across iterative repair runs, grounding the motivation directly in the fine-tuning regime used in the experiments.
2. **Differentiation from Targeted Active Learning and Submodular Subset Selection**
   - *Context:* In Section 2 (lines 095–102), the text cites targeted active learning for rare classes and slices (TALISMAN; Kothawade et al., 2022) and data subset selection under distribution shift (ORIENT; Karanam et al., 2022). It differentiates the proposed setting by stating: *"Our setting differs from standard AL because the goal is not generic sample efficiency, but selecting source data that complements a targeted repair set and corrects an attribute-defined failure slice."*
   - *Potential Concern:* While standard AL focuses on generic sample efficiency, targeted AL frameworks (e.g., TALISMAN, SIMILAR) and distribution-shift subset selection (ORIENT) are specifically designed to target rare failure slices or distribution shifts using Submodular Mutual Information (SMI) conditioned on a target query set. In model repair, the failure slice exemplars or the generated repair set D_{gen} can serve as the target query. The text dismisses these methods as generic AL without explaining why query-guided SMI is insufficient or conceptually distinct from AttriSelect's count-based rarity and penalty formulation.
   - *Suggested Improvement:* Clarify how the repair-time source-selection problem differs operationally from query-guided targeted AL. Specifically, discuss that: (1) AttriSelect explicitly accounts for synthetic redundancy via the generated-data frequency penalty ($P_{i,a}$), which SMI functions do not penalize; and (2) AttriSelect uses closed-form categorical counts rather than computing continuous kernel similarity matrices that scale quadratically with candidate pool size.
3. **Formulation Gap Between Image-Level Attributes and Multi-Instance Dense Prediction Tasks**
   - *Context:* In Section 3 (lines 124–129), the mathematical framework defines attributes at the image level: *"let each image $x \in \mathcal{X}$ have attributes $A = a_1, \ldots, a_m$. Each attribute $a_j : \mathcal{X} \to V_j$ maps an image to a discrete value, such as a color, weather condition, texture, or viewpoint. A slice defined by several attribute values is $T = x \mid a_{j_1}(x) = v_{j_1}, \ldots, a_{j_k}(x) = v_{j_k}$."*
   - *Potential Concern:* This formulation assumes a single scalar value per attribute field for each image x. While this fits single-label classification (ImageNet10, CelebA), it is ill-defined for multi-instance dense prediction tasks evaluated in the paper (COCO keypoint estimation and KITTI object detection). In scenes with multiple people or vehicles, individual instances frequently possess conflicting attributes (e.g., different clothing colors, occlusions, or poses). The main text does not specify how instance-level attributes are aggregated into an image-level tuple, nor how class-conditional counts $n_a(v \mid c)$ operate when an image contains instances from multiple classes.

- *Suggested Improvement:* Formalize in Section 3 how multi-instance images are handled. If the upstream VLM queries the "main target object" (as indicated in Appendix D, line 876) or aggregates instance attributes, state this reduction explicitly in the problem formulation.

4. **Notational Ambiguity in the Failure Slice Definition (Equation 1)**
   - *Context:* In Section 4.1 (lines 157–159), Equation (1) defines an attribute failure slice T_{bug}:

     $$|T_{\text{bug}} \cap D_{\text{train}}| < \rho |D_{\text{train}}|, \quad \text{Acc}(T_{\text{bug}}) < \text{Acc}(D_{\text{val}})$$

   - *Potential Concern:* In Section 3 (lines 125–127), a slice T is defined over the general input space $\mathcal{X}$. In Equation (1), the notation $\text{Acc}(T_{\text{bug}})$ does not specify the evaluation dataset on which slice accuracy is measured. Because lines 155–156 state that the repair pipeline analyzes the validation behavior of f_{θ_t}, the condition evaluates validation examples matching the slice description. As written, $\text{Acc}(T_{\text{bug}})$ leaves the exact evaluation split ambiguous.
   - *Suggested Improvement:* Update Equation (1) to explicitly define the evaluation split, for example: $\text{Acc}(T_{\text{bug}} \cap D_{\text{val}}) < \text{Acc}(D_{\text{val}})$, or formally define $\text{Acc}_{D}(T_{\text{bug}})$.

Minor Corrections and Typos

- **Line 011, 015, 018, 023:** In the PDF text conversion of the Abstract, hyphenated line breaks introduce unicode or spacing artifacts (e.g., `corresponding` on line 011, `repairrelevant` on line 01 `coverage` on line 018, `classification` on line 023). Ensure that standard LaTeX hyphenation ligatures are properly handled.
- **Line 126–127:** The definition $T = x \mid a_{j_1}(x) = v_{j_1}, \ldots, a_{j_k}(x) = v_{j_k}$ uses double subscripting a_{j_1}, whereas a_j is defined immediately prior as mapping to V_j. Using consistent indexing (e.g., $a_1(x) = v_1, \ldots, a_k(x) = v_k$ or $a_j(x) = v_j$) would improve clarity.
- **Line 131–132:** The text defines the source pool as $D_{\text{src}} \subseteq D_{\text{train}}$. In standard setups $D_{\text{src}} = D_{\text{train}}$. Clarifying briefly why D_{src} is a strict subset (e.g., subsampling for the COCO benchmark) would prevent confusion regarding data splits.

[AttriSelect Selection Framework and Algorithmic Design] (Pages: 3-6, 14-17)

Summary

This segment comprises the core methodology of AttriSelect (Section 4, pages 3–6), accompanied by its implementation details (Appendix B, pages 14–15), computational cost accounting (Appendix B.1, pages 15–16), algorithmic pseudocode (Appendix C, Algorithm 1, page 16), and attribute assignment protocols (Appendix D, pages 16–17). AttriSelect is formulated as a repair-time source-data selection mechanism situated between failure-slice diagnosis/synthetic data generation and model retraining. Given an upstream diagnosed failure slice T_{bug}, an annotated source pool $\mathcal{D}_{\text{src}}$, a budget n_{select}, and a fixed set of synthetic repair images $\mathcal{D}_{\text{gen}}$, $AttriSelect selects a compact source subset \mathcal{D}_{\text{sel}} \subseteq \mathcal{D}_{\text{src}}$ to accompany the synthetic data during retraining.

The candidate selection objective combines three terms:

1. **Class-conditional marginal rarity ($R_{i,a}$)**, which prioritizes source instances possessing attribute values that occur infrequently within their respective class;
2. **Generated-data frequency penalty ($P_{i,a}$)**, which penalizes attribute values that are heavily represented in the combined source and synthetic pool to mitigate redundancy;
3. **Soft class balancing ($C_c^{(k)}$)**, which applies a dynamic additive correction during greedy selection to prevent budget concentration in a small number of classes.

The segment also details the use of vision-language models (GPT-5 to propose candidate attribute taxonomies and Qwen3-VL to annotate dataset images via visual question answering) and provide wall-clock profiling that distinguishes the one-time offline VLM annotation pass from online subset selection.

Potential Mistakes and Improvements

1. **Undefined Class Assignment (y_i) and Class Balancing ($C_c^{(k)}$) for Dense Perception Tasks (COCO and KITTI)**
   - *Description & Potential Concern:* Equations (2) and (5), as well as Algorithm 1 (lines 1, 11, and 16), explicitly require each candidate image x_i to possess a single discrete class label $y_i \in 1, \ldots, N_{\text{cls}}$ to compute class-conditional counts $n_a(v_{i,a} \mid y_i)$ and soft class-balancing bonuses $C_c^{(k)} = w \cdot (t - n_c^{(k)})/t$, where $t = n_{\text{select}}/N_{\text{cls}}$. While y_i is standard for single-label image classification (ImageNet10, CelebA), its operational definition is ambiguous for dense prediction benchmarks:
     - In COCO person keypoint estimation, the task targets a single object category (`person`), which implies $N_{\text{cls}} = 1$. If $N_{\text{cls}} = 1$, then $t = n_{\text{select}}$, and at greedy selection step k, the class counter is $n_1^{(k)} = k$. The class-balancing bonus $C_1^{(k)} = w \cdot (t - k)/t$ then evaluates to an identical scalar constant across all remaining candidates at step k. Adding an identical scalar constant to every candidate cannot alter the greedy ranking ($\arg\max_i (\bar{s}_i + C^{(k)}) = \arg\max_i \bar{s}_i$), rendering $C_c^{(k)}$ mathematically incapable of changing subset composition. Howeve Table 33 sets $w = 15$ for COCO, Table 35 reports performance variations as w changes (e.g., AP is 0.6148 at $w = 0$ vs. 0.6183 at $w = 15$ at ratio 0.1), and Table 36 demonstrates distinct metrics when ablating C ($R + P + C$ achieves 0.6183 vs. 0.5954 for $R + P$). In addition, Table 1 shows that `STRATIFIED` sampling differs substantially from `RANDOM` on COCO (8.32 vs 3.37 RER at ratio 0.3). This indicates that COCO candidate images were partitioned into non-trivial sub-classes or pseudo-groups, but this partitioning is not defined in the text.
     - In KITTI object detection, individual road scenes frequently contain multiple co-occurring instances of different categories (e.g., both `Car` and `Pedestrian` in the same frame). The tex does not explain how multi-instance images are mapped to a single scalar class label y_i for rarity conditioning, nor how class quotas $n_c^{(k)}$ are updated upon selecting a multi-object image.
   - *Suggested Improvement:* Document explicitly how y_i, N_{cls}, and the class allocation counters $n_c^{(k)}$ are defined for COCO keypoint estimation (e.g., whether candidates are partitioned by pose clusters, instance counts, or scene types) and for multi-instance KITTI scenes (e.g., primary instance category, rarest detected category, or multi-label quota tracking).
2. **Ambiguity in Attribute Set $\mathcal{A}$ Between Task-Level Vocabulary and Active Repair Fields**
   - *Description & Potential Concern:* In Section 4.1 (line 124) and Table 5 (line 712), $\mathcal{A} = a_1, \ldots, a_m$ is formally defined as the complete task-level attribute vocabulary proposed by the VLM (e.g., comprising 10 categories for KITTI as listed in Figure 3). In Equation (4) and Algorithm 1 (line 6), the base score $\bar{s}_i$ is defined as summing over all $a \in \mathcal{A}$. Conversely, Section 4.2 (line 231) describes the base score as summing "over the repair attributes", lines 764–765 and 868–870 state that for KITTI only `color` and `weather` are used as repair fields while the remaining categories are retained solely for diagnostics, and Appendix J.3 refers to the main setups as "two-field selector[s]".
   - *Impact on Reproducibility:* If an implementer follows Algorithm 1 literally by summing over all $a \in \mathcal{A}$, candidate scoring will incorporate non-repair attributes (e.g., `viewpoint` , `occlusion` , `motion_state`), yielding candidate rankings that conflict with the experimental results.
   - *Suggested Improvement:* Introduce distinct notation for the active failure-slice fields (e.g., $\mathcal{A}_{\text{repair}} \subseteq \mathcal{A}$ or $\mathcal{A}_{T}$) $and specify in Equation (4) and Algorithm 1 that the summation is evaluated over a \in \mathcal{A}_{\text{repair}}$ during targeted repair.
3. **Unspecified Scaling Formulation for Generated Pool Attribute Counts ($\tilde{n}_a^{\text{gen}}$)**
   - *Description & Potential Concern:* In Section 4.2 (lines 224–226) and Algorithm 1 (line 2), the combined effective count is written as $\hat{n}_a(v) = n_a^{\text{src}}(v) + \tilde{n}_a^{\text{gen}}(v)$, where $\tilde{n}_a^{\text{gen}}(v)$ denotes "the generated count used by the selector after any scaling of the generated pool." However, neither Section 4, Algorithm 1, nor the hyperparameter specifications in Appendix M (Table 33) describe what scaling factor or function is applied.
   - *Impact on Soundness & Clarity:* The penalty term $P_{i,a}$ in Equation (3) depends directly on the relative magnitude of synthetic attribute counts versus source counts. If an unstated scaling factor γ was applied (e.g., $\tilde{n}_a^{\text{gen}}(v) = \gamma \cdot n_a^{\text{gen}}(v)$) to balance the fixed 1,000 generated images against source pools varying from 5,000 to 80,000 images, this parameter must be documented. If no scaling was applied, the text should clarify that unscaled raw integer counts were used.
   - *Suggested Improvement:* State explicitly whether $\tilde{n}_a^{\text{gen}}(v) = n_a^{\text{gen}}(v)$ across all experiments, or provide the exact formula and numerical value used for the scaling parameter.
4. **Algorithmic Formulation and Complexity of Algorithm 1 vs. Empirical Runtimes**
   - *Description & Potential Concern:* In Algorithm 1 (lines 10–17), the soft class-balance correction C_c and adjusted score $\tilde{s}_i are represented inside a loop over every unselected candidate \in \mathcal{D}_{\text{src}} \setminus \mathcal{D}_{\text{sel}}^{\text{src}} at each selection step k \in 1, \ldots, n_{\text{select}}$. A literal implementation of this nested structure requires $O(n_{\text{select}} \cdot |\mathcal{D}_{\text{src}}| \cdot |\mathcal{D}_{\text{sel}}|)$ candidate score updates, amounting to approximately 2.4 \times 10^9 inner-loop iterations on CelebA at ratio 0.5 ($|\mathcal{D}_{\text{src}}| = 80{,}000$, $n_{\text{select}} = 40{,}000$).
   - *Impact on Clarity:* As established in Proposition 5, the class bonus $C_c^{(k)}$ is identical for all candidates within class c at step k, meaning the intra-class candidate ranking induced by $\bar{s}_i$ is strictly invariant throughout selection. The sub-second execution times reported in Table 2 (e.g., 1.37 seconds on CelebA at ratio 0.5) indicate that the actual implementation

N{\text{cls}}$pre − sortedpriorityqueuesevaluatedin$O(|\mathcal{D}{\text{src}}| \log |\mathcal{D}{\text{src}}| + n_{\text{select}} \log N_{\text{cls}})$ time.

- *Suggested Improvement:* Add an explanatory remark to Appendix C clarifying this implementation detail, reconciling the conceptual while-loop in Algorithm 1 with the empirical runtimes reported in Table 2.

5. **Spatial Conditioning and Query Protocol for VLM Object Attributes**
   - *Description & Potential Concern:* Appendix D (Figure 3) provides prompt templates for KITTI attribute assignment via Qwen3-VL, such as `"Prompt: a #1 main target object in a driving scene"` for object color, scale, viewpoint, occlusion, and truncation. In driving scenes containing multiple vehicles and pedestrians at varying positions and depths, the text does not describ how the VLM identifies which entity constitutes the "main target object". Specifically, it is not specified whether ground-truth bounding box crops are provided to the model, whether spatia coordinates/prompts are injected into the query, or whether full uncropped images are evaluated directly.
   - *Suggested Improvement:* Clarify the visual input format provided to Qwen3-VL for object-level versus scene-level attributes in Appendix D, and describe how attribute assignments are resolved when multiple instances appear in a single frame.

Minor Corrections and Typos

1. **Table 5 Definition of $\hat{n}_a(v)$ (page 14, line 717):** The glossary defines $\hat{n}_a(v)$ as *"Frequency count of attribute value v in the generated repair pool"*. This contradicts Section 4.2 (line 224) and Algorithm 1 (line 2), where $\hat{n}_a(v) = n_a^{\text{src}}(v) + \tilde{n}_a^{\text{gen}}(v)$ represents the combined effective count across both source and generated pools, while the generated count alone is denoted $\tilde{n}_a^{\text{gen}}(v)$. Table 5 should be updated to reflect the combined count.
2. **Algorithm 1 Notation for Set Difference (page 16, lines 844 and 848):** In lines 11 and 15 of Algorithm 1, the set difference is rendered as `x_i \in D/src sel` and `\arg\max_{x_i \in D/src sel}` . This should be typeset using standard set subtraction notation, e.g., $x_i \in \mathcal{D}{\text{src}} \setminus \mathcal{D}{\text{sel}}^{\text{src}}$.
3. **Table 6 vs. Figure 3 Attribute Count for KITTI:** Figure 3 lists 10 attribute categories for KITTI (`color` , `object_scale` , `viewpoint` , `occlusion` , `truncation` , `motion_state` , `background_style` , `traffic_density` , `lighting_condition` , `weather`), whereas Table 6 reports that 6 attributes were assigned by Qwen3-VL. Specifying in Appendix D which 6 categories were retained for source-pool annotation would resolve the discrepancy.
4. **Table 33 Missing Parameter ε:** The numerical stability constant $\varepsilon > 0$ introduced in Equation (2) and used in Algorithm 1 is omitted from the hyperparameter specifications in Table 33. The numerical value used in the experiments should be explicitly stated.
5. **Equation (2) vs. Algorithm 1 Line 4 Exponent Formatting:** In Equation (2) (page 4), the rarity term is formatted as $(1/(n_a(v_{i,a} \mid y_i) + \varepsilon))^{\alpha}$, whereas in Algorithm 1 (line 4, page 16) it is writte as $(n_a(v_{i,a} \mid y_i) + \varepsilon)^{-\alpha}$. Adopting uniform notation across both sections would improve visual consistency.

[Theoretical Analysis and Mathematical Guarantees] (Pages: 5-6, 26-32)

Summary

This segment comprises Section 4.3 (pages 5–6) and Appendix L (pages 26–32), which formulate the theoretical foundations and mathematical guarantees for the AttriSelect framework. The theoretical analysis establishes the limitations of attribute-agnostic active learning (AL) selectors and characterizes the conditions under which attribute-driven source selection enriches rare semantic failure slices while maintaining class balance:

1. **Base-Rate Limits of Attribute-Agnostic Active Learning (Proposition 1):** Proves that if candidate non-attribute selection signals (such as model loss, predictive entropy, feature embeddings, detector difficulty) are conditionally independent of failure-slice membership given class c, any selector choosing B_c samples from class c selects target-slice instances in expectation strictly in proportion to their class-conditional prevalence, $B_c \frac{|T_c|}{N_c}$.
2. **Geometric Coverage Limits (Equations 10–12):** Analyzes k-center core-set selection via empirical covering numbers ($\mathcal{N}_\epsilon^{\text{emp}}$), showing that while a compact target slice can be geometrically covered in isolation by a small empirical covering number q_c, this isolated covering bound does not constrain or guarantee target sample density in a global fixed-budget solution.
3. **Rarity-Dominates-Penalty Regime (Propositions 2 and 3):** Defines the pairwise rarity advantage A_{ij} and frequency-penalty disadvantage F_{ij} between target and non-target candidates in cla c. It establishes an upper threshold $\lambda_c^\star = \min_{x_i \in T_c, x_j \notin T_c, F_{ij} > 0} \frac{A_{ij}}{F_{ij}}$ such that for $0 \le \lambda < \lambda_c^\star$, every target instance in T_c achieves a strictly higher base score than every non-target instance in that class, yielding a deterministic target-slice enrichment guarantee of $\min B_c, |T_c|$ samples.
4. **Heuristic Failure Modes and Boundary Settings (Propositions 4, 7, and 8):** Proves that uncertainty sampling selects zero target-slice instances if non-target candidate uncertainties exceed all target uncertainties (Proposition 4), extends deterministic slice enrichment to the zero-generation setting $D_{\text{gen}} = \emptyset$ (Proposition 7), and notes the impossibility of source-only repair when a failure slice has zero support in the source pool ($T \cap D_{\text{src}} = \emptyset$, Proposition 8).
5. **Class Imbalance Dynamics (Propositions 5 and 6):** Proves that an infinite balancing weight ($w \to \infty$) recovers round-robin class quotas differing across active classes by at most one sample, and bounds instantaneous step-wise class count disparities by $\frac{t\Delta}{w}$ for finite balancing weights $w \in (0, \infty)$.
6. **Surrogate Repair Utility (Equation 21):** Formalizes expected repair utility as an affine function of target-slice representation.

Potential Mistakes and Improvements

1. **Theoretical and Operational Tension in the Margin Condition ($\lambda < \lambda_c^\star$) of Propositions 2 and 3**
   - *Potential Concern:* Proposition 2 establishes that target-slice instances outrank all non-target instances in class c under the condition $\lambda < \lambda_c^\star = \min_{x_i \in T_c, x_j \notin T_c, F_{ij} > 0} \frac{A_{ij}}{F_{ij}}$, which Proposition 3 directly adopts to establish the deterministic lower bound of $\min B_c, |T_c|$ target samples. However, there is a conceptual divergence between this mathematical premise and the algorithmi motivation for the frequency penalty $P_{i,a}$ detailed in Section 4.2 (lines 218–226):
     - The penalty $P_{i,a}$ is explicitly designed to downweight target attribute values once they are heavily supplied by the synthetic repair pool D_{gen}, allowing the selector to reserve remaining source budget for complementary, non-target contexts.
     - When $\lambda < \lambda_c^\star$, the frequency penalty is mathematically constrained never to overturn the rarity advantage of any target instance over any non-target instance in the same class. In this regime, the selector must exhaust all $|T_c|$ real target instances before selecting a single non-target instance from class c, preventing the penalty from prioritizing complementary non-target samples over target samples.
     - Conversely, when $\lambda \ge \lambda_c^\star$, the penalty can downweight target instances below non-target instances. However, in this operational regime, the premise of Proposition 2 is violated, and Proposition 3 no longer guarantees the deterministic bound $\min B_c, |T_c|$.
     - In the empirical configurations (Table 33), the hyperparameters set $\alpha \in 0.01, 0.03$ and $\lambda \in 0.5, 5.0$. Because α is small ($n^{-\alpha} \approx 1 - \alpha \ln n$, making $A_{ij} \ll 0.1$) and the addition of 1,000 synthetic repair images creates a substantial positive penalty gap ($F_{ij} > 0$), $\lambda_c^\star$ evaluates to a small fraction. In practical settings such as KITTI ($\lambda = 5.0, \alpha = 0.01$), the algorithm operates in the regime $\lambda \ge \lambda_c^\star$.
   - *Suggested Improvement:* Delineate two operating regimes: (i) the *target-preserving regime* ($0 \le \lambda < \lambda_c^\star$), where real target examples take absolute priority over synthetic redundancy penalties, and (ii) the *redundancy-suppression regime* ($\lambda \ge \lambda_c^\star$), where the selector intentionally allows diverse non-target examples to outscore redundant target examples. Acknowledging that empirical configurations operate in the latter regime would clarify the relationship between Proposition 3 and the experimental results.
2. **Probability Space Specification and Joint Independence in Proposition 1**
   - *Potential Concern:* In Proposition 1 (lines 277–279 and 1426–1428), the conditional independence assumption is stated marginally for individual samples: *"If the target-slice indicator $\mathbf{1}[x_i \in T$ is conditionally independent of h_i given class c..."* However, the proof (lines 1438–1446) requires that $I_i = \mathbf{1}[x_i \in T_c]$ be conditionally independent of the *entire joint candidate collection* $({h_j} *{j=1}^{N_c}, U) given class c in order to evaluate* $\mathbb{E}[Z_i \mid h_j, U] = Z_i \mathbb{E}[I_i]$. *Because active learning selection indicators* $Z_i = \mathbf{1}[x_i \in D$ {\text{AL}}} $depend on sorting, clustering, or batch greedy selection overall candidate signals* {h_1, \dots, h_{N_c}}, *marginal pairwise independence* (I_i \perp h_i \mid y_i=c)*does not imply joint conditional independence* (I_i \perp {h_j}_{j=1}^{N_c} \mid c$) without an assumption of mutual independence or exchangeability across samples.
   - *Probability Space Formulation:* In lines 270–272, D_{src} and T_c are defined as fixed, finite sets. For a fixed sample pool and fixed candidate index i, the indicator $I_i = \mathbf{1}[x_i \in T_c]$ is a deterministic constant in $0, 1$, so $\mathbb{E}[I_i]$ evaluates to $I_i \in 0, 1$ rather than the base rate $|T_c|/N_c$.
   - *Suggested Improvement:* Formalize the underlying probability space—such as defining candidate indices as an exchangeable uniform random permutation of the pool, or modeling candidates as conditionally exchangeable draws given $y = c$. In addition, specify the premise as joint conditional independence: $(I_1, \dots, I_{N_c}) \perp (h_{j=1}^{N_c}, U) \mid c$.
3. **Restrictiveness of $A_{ij} > 0$ Across Multi-Attribute Vocabularies in Proposition 2**
   - *Potential Concern:* Proposition 2 assumes that $A_{ij} > 0$ for *every* target/non-target pair (x_i, x_j) within class c, where:

$$A_{ij} = \sum_{a \in D_{ij}} \left[\left(\frac{1}{n_a(v_{i,a} \mid c) + \varepsilon} \right)^{\alpha} - \left(\frac{1}{n_a(v_{j,a} \mid c) + \varepsilon} \right)^{\alpha} \right]$$

In Eq. (4), the base score $\bar{s}is a sum s rarity terms over the attribute vocabulary \mathcal{A}. When \mathcal{A} includes auxiliary attribute fields beyond those defining the target failure slice (e.g., in KITTI where \mathcal{A} contains 10 attributes including viewpoint, lighting, and occlusion, while Tis defined only on color and weather), a non-target sample x_j \notin T_c may carry a natypical value on an auxiliary field. If the rarity contribution of x_j on that auxiliary field exceeds the target rarity advantage on the slice fields,A_{ij} becomes negative. If A_{ij} \le 0 for any pair where F_{ij} > 0, the ratio \frac{A_{ij}}{F_{ij}} is non-positive, rendering \lambda_c^\star \le 0 and collapsing the valid parameter interval [\lambda_c^\star)$.

- *Suggested Improvement:* Clarify whether the summation in Eq. (4) and Proposition 2 is evaluated over the diagnosed repair fields $\mathcal{A}_{\text{slice}} \subseteq \mathcal{A}$ or over the complete task vocabulary. If evaluate over the full vocabulary, discuss the condition that incidental attribute rarity does not overpower the target-defining rarity advantage.

4. **Dynamic Step-Dependency and Budget Scaling of the Disparity Bound in Proposition 6**
   - *Potential Concern:* In Proposition 6 (Eq. 22), the score spread is defined at greedy step k as $\Delta = \max_c A_c^{(k)} - \min_c A_c^{(k)}$, where $A_c^{(k)}$ denotes the highest remaining base score in class c.
     - Because candidates are consumed greedily, $A_c^{(k)}$ varies across iterations, making Δ step-dependent ($\Delta^{(k)}$). Notationally omitting the index k in Eq. (23) could suggest that Δ is a static dataset constant.
     - In line 1603, the phrase *"assume all active class-head scores lie in an interval of width* $\Delta = \max_c A_c^{(k)} - \min_c A_c^{(k)}$*"* is tautological, as any finite set of scalars lies in the range spanned by its maximum and minimum.
     - Furthermore, the bound $n_d^{(k)} - n_c^{(k)} \le \frac{t\Delta^{(k)}}{w}$ applies immediately *prior* to step k. After candidate selection, the count of class d increments by 1, yielding an instantaneous post-selectio gap of $n_d^{(k+1)} - n_c^{(k)} \le \frac{t\Delta^{(k)}}{w} + 1$.
     - Because $t = n_{\text{select}}/N_{\text{cls}}$, the bound scales linearly with the selection budget n_{select}, bounding the relative allocation disparity $\frac{n_d^{(k)} - n_c^{(k)}}{t} \le \frac{\Delta^{(k)}}{w}$ rather than enforcing a constant absolute sample count ceiling.
   - *Suggested Improvement:* Index the spread parameter explicitly as $\Delta^{(k)}$ in Equations (22) and (23), or define a uniform global bound $\Delta_{\max} = \max_{x \in D_{\text{src}}} \bar{s}(x) - \min_{x \in D_{\text{src}}} \bar{s}(x)$. State the terminal bound across active classes as $\max_{c,d} |n_d - n_c| \le \lfloor \frac{t\Delta_{\max}}{w} \rfloor + 1$.
5. **Formal Framing of Proposition 8**
   - *Potential Concern:* Proposition 8 ("Impossibility for absent source slices") states that if $D_{\text{gen}} = \emptyset$ and $T \cap D_{\text{src}} = \emptyset$, then every subset $S \subseteq D_{\text{src}}$ satisfies $|S \cap T| = 0$. This statement is an immediate consequence of elementary set theory ($S \subseteq D_{\text{src}} \implies S \cap T \subseteq D_{\text{src}} \cap T = \emptyset$). Presenting this definitional property as a formal numbered proposition with a proof environment elevates an elementary set relation to the level of substantive theorems.
   - *Suggested Improvement:* Recast Proposition 8 as an informal "Remark" or introductory observation in Section L.1.

Minor Corrections and Typos

1. **Exact Cardinality in Proposition 3 and Proposition 7:** Propositions 3 and 7 state that AttriSelect includes *"at least* $\min B_c, |T_c|$*"* target examples from class c. Under the ranking condition of Proposition 2, target candidates strictly precede non-target candidates. Because class c contains only $|T_c|$ target examples and exactly B_c samples from class c are selected, the subset cannot contain more than $\min B_c, |T_c|$ target samples. The cardinality holds with exact equality: $|D_{\text{sel}}^{\text{src}} \cap T_c| = \min B_c, |T_c|$.
2. **Forward Reference to Proposition 2 (Line 200):** In Section 4.2 (page 4), the text states: *"the conditional guarantee is stated in Proposition 2."* Because Proposition 2 appears in Appendix L (page 28) and does not appear in the main text (which contains only Proposition 1), the citation should specify *"Proposition 2 in Appendix L"*.
3. **Notation Overloading for Budget (m vs. n_{select}):** On page 6 (line 270), $m = \lfloor rN \rfloor$ denotes the total source selection budget. On page 3 (line 124), m denotes the attribute vocabulary size ($\mathcal{A} = a_1, \ldots, a_m$). Standardizing the selection budget to n_{select} (as used in Section 4.2, Table 5, and Algorithm 1) would avoid ambiguity.
4. **Notation Overloading on A in Proposition 6:** The symbol A is used concurrently for the attribute vocabulary $\mathcal{A}$ (line 124), the rarity advantage A_{ij} (Eq. 14), and the class-head base score $A_c^{(k)}$ (Eq. 22). Renaming $A_c^{(k)}$ to $M_c^{(k)}$ or $h_c^{(k)}$ would resolve the symbol collision.
5. **Mismatched Outer Delimiters in Equation (14):** In Appendix L (line 1483, Eq. 14), the first term opens with an outer bracket [but closes with an outer parenthesis) before the minus sign. The delimiters should be matched consistently: $\left[\left(\frac{1}{n_a(v_{i,a}|c)+\varepsilon} \right)^{\alpha} - \left(\frac{1}{n_a(v_{j,a}|c)+\varepsilon} \right)^{\alpha} \right]$.
6. **Set Membership Notation in Equation (16):** The subscript in Eq. (16) writes $x_j \in /T_c$ using a slash. Using standard set exclusion notation $x_j \notin T_c$ would be cleaner.
7. **Self-Referential Proof Citation in Appendix L (Line 1662):** The text states, *"The proof is identical to the rare-slice enrichment proof in Appendix L..."* Because this text appears within Appendix L, it should cite the proof of Proposition 3 directly.

[Main Repair Evaluation, Efficiency, and Core Ablations] (Pages: 6-12)

Summary

This segment comprises the empirical evaluation (Section 5), conclusion (Section 6), formal compliance statements (AI use, ethics, reproducibility), and initial reference list (pages 6–12), with empiric ties to Appendices B, E, F, H, I, J, and N. The experimental section benchmarks the proposed attribute-driven data selection framework, AttriSelect, across four vision benchmarks spanning three tas modalities:

1. **Multi-class Image Classification**: A 10-class ImageNet subset (ImageNet10; 10,400 training images) evaluated on ResNet-18;
2. **Binary Attribute Classification**: CelebA lipstick-wearing classification (80,000 training images) evaluated on ResNet-18;
3. **Multi-Person Pose Estimation**: MS-COCO person keypoint detection (20,000 training images) evaluated using Keypoint R-CNN with a ResNet-50-FPN backbone;
4. **2D Object Detection**: KITTI Car and Pedestrian detection (5,000 training images) evaluated with YOLOv8n.

AttriSelect combines class-conditional marginal rarity (R), a generated-data frequency penalty (P), and soft class balancing (C) to select a compact subset of real source data ($D_{\text{sel}}^{\text{src}}$) to accompany 1,000 diffusion-generated repair images ($D_{\text{sel}}^{\text{gen}}$). The primary evaluation tracks Relative Error Reduction (RER, Eq. 7) across source data retention ratios $r \in 0.1, 0.3, 0.5, 0.7, 0.9$ alongside unselecte full-source retraining ($r = 1.0$). AttriSelect is compared against seven generic active learning (AL) and data-selection baselines (Random, Stratified, Entropy, CoreSet, BADGE, PPAL, and Uncertainty Herding). The segment also reports online subset selection runtimes on COCO and CelebA (Table 2), marginal attribute accuracy difference heatmaps on ImageNet10 (Figure 2), COCO failure-slice repair counts (Table 3), and an ablation of the three scoring components on ImageNet10 (Table 4).

Potential Mistakes and Improvements

1. **Reporting Overall Benchmark Test Metrics as Headline Results in Table 1 Masks Target-Slice Repair Dynamics**
   - *Description & Context:* The core objective of targeted vision model repair is to remediate systematic model failures on diagnosed rare semantic slices while preserving general model competence. However, in Table 1, the headline metric reported across all budget ratios is Relative Error Reduction (Eq. 7) computed exclusively over the *entire unsegmented test set* ($M_{\text{before}} = 0.7173$ on ImageNet10, 0.9057 on CelebA, 0.5902 on COCO, and 0.5626 on KITTI). Within these test splits, the diagnosed failure slice forms a very small fraction of the evaluatic data (e.g., $n = 16$ out of 1,300 test images in ImageNet10, or ≈ 1.2; $n = 22$ out of 1,173 in COCO, or ≈ 1.8; and only 2 test images per seed in KITTI, as noted in lines 1026–1027).
   - *Potential Concern:* In Appendix F.3 (Table 17, page 20), on the exact ImageNet10 failure conjunction ("pink AND fabric", $n = 16$ at ratio 0.5), AttriSelect achieves an accuracy of 0.7625 ± 0.0523. In comparison, simpler heuristic controls achieve higher raw accuracy on the failure slice itself (Target-Slice + Stratified achieves 0.7750 ± 0.0559, and Attribute-Filtered Random achieves 0.7750 ± 0.0342). AttriSelect's higher overall RER in Table 1 is driven substantially by preserving performance on the non-target population ($n = 1,284$, test accuracy 0.7507 vs. $0.7435/0.7438$) through its soft class-balancing term (C). Confining slice-specific metrics to marginal attribute deltas at a single budget (Figure 2) and COCO slice counts (Table 3 leaves it ambiguous whether the headline RER advantages across all budgets reflect superior repair of the diagnosed failure mode or general regularization and class balancing across nominal classes.
   - *Suggested Improvement:* Report target-slice metrics (such as target-slice RER, worst-group accuracy, or slice keypoint AP) directly alongside overall RER in Table 1 across all source retention ratios, or provide a dedicated summary table in Section 5.1 tracking target-slice recovery across tasks.
2. **Undefined Operational Threshold for "Repaired Slices" in Table 3 (Section 5.4)**

- *Description & Context:* Section 5.4 (lines 425–431) and Table 3 report the count of repaired COCO failure slices out of 12 candidate slices across selection ratios (e.g., stating that AttriSelect repairs 10 slices at ratio 0.1 compared to 7 for the best active learning baseline, and reaches 11 repaired slices at ratios 0.5–0.9).
- *Potential Concern:* Neither Section 5.4, Table 3, nor the surrounding text provides an explicit mathematical condition or threshold defining a slice as "repaired." Equation 1 formally defines a failure slice at diagnosis time ($|T_{\text{bug}} \cap \mathcal{D}_{\text{train}}| < \rho \, |\mathcal{D}_{\text{train}}| \, and \, \text{Acc}(T_{\text{bug}}) < \text{Acc}(\mathcal{D}_{\text{val}})$), but it is unspecified whether "repaired" indicates that post-repair slice AP reaches the validation set average ($\text{AP}_{\text{after}}(T_{\text{bug}}) \ge \text{AP}_{\text{after}}(\mathcal{D}_{\text{val}})$), achieves a strictly positive relative error reduction ($\text{RER} > 0$), exceeds the pre-repair checkpoint by a non-zero margin, or meets a specific statistical significance criterion. Without an operational definition, these counts cannot be independently replicated or verified.
- *Suggested Improvement:* Explicitly state the exact mathematical condition and evaluation threshold used to designate an error slice as "repaired" in the text of Section 5.4 and in the caption of Table 3.

3. **Asymmetry in Reporting Selection Latency Speedups vs. Upfront VLM Preprocessing Overhead (Table 2 vs. Tables 6–8)**
   - *Description & Context:* Section 5.2 states that AttriSelect is "cheaper than generic active-learning selectors" (line 366) and reports in Table 2 that it selects subsets in under two seconds, yielding speedups of "at least $39\times$ faster on CelebA and more than $230\times$ faster on COCO" compared to non-attribute active learning baselines.
   - *Potential Concern:* Table 2 measures only the greedy candidate sorting loop *after* all source images have already been annotated with discrete attribute records. In contrast, standard active learning baselines (e.g., Entropy, PPAL, Uncertainty Herding) compute acquisition scores directly from model forward passes or gradients without requiring auxiliary vision-language model annotations. As documented in Appendix B.1 (Table 6 and Table 8), tagging the source pool using Qwen3-VL on an NVIDIA H200 GPU requires 29.35 minutes for ImageNet10 (10,400 images), 32.77 minutes for KITTI (5,000 images), and 90.00 minutes for COCO (20,000 images). Consequently, in a cold-start first-repair setting (Appendix Table 8, COCO ratio 0.1), AttriSelect's total wall-clock time is 1.57 hours (94.2 minutes), whereas PPAL requires 6.97 minutes and Uncertainty Herding requires 5.08 minutes. While lines 376–377 state that these are subset-selection speedups rather than end-to-end pipeline speedups, describing the method as strictly faster in the main text without presenting the upfront VLM tagging cost alongside Table 2 presents an incomplete operational picture for single-run or cold-start deployments.
   - *Suggested Improvement:* Reference the one-time VLM annotation overhead from Table 6 directly in Section 5.2 alongside Table 2, and explicitly state the operational break-even point (in terms of repeated repair cycles or slice queries on the same corpus) where online ranking speed amortizes the initial VLM indexing overhead.
4. **Benchmarking Exclusively Against Slice-Agnostic Active Learning Baselines in the Primary Benchmark**
   - *Description & Context:* In Table 1, AttriSelect is compared exclusively against generic, slice-agnostic active learning methods (Random, Stratified, Entropy, CoreSet, BADGE, PPAL, and Uncertainty Herding). These methods do not receive information regarding the diagnosed failure slice or attribute metadata.
   - *Potential Concern:* By Proposition 1, any selector whose acquisition variables are conditionally independent of the failure slice cannot systematically enrich the target slice beyond its natural base-rate prevalence. Outperforming slice-blind baselines confirms the premise of Proposition 1, but does not establish whether the specific $R + P + C$ formulation improves upon existing targeted active learning or submodular subset selection algorithms specifically designed to handle rare classes, slices, and distribution shifts (such as TALISMAN [Kothawade et al., ECCV 2022] for object detection and ORIENT [Karanam et al., NeurIPS 2022], both cited in lines 096–099). While Appendix F provides ad-hoc heuristic controls (Target+Stratified, Filtered Random, Filtered Entropy, and Generated-Complement), omitting established targeted active learning baselines leaves it unproven whether the proposed three-term scoring function improves upon existing query-guided or submodular targeted subset selection algorithms.
   - *Suggested Improvement:* Benchmark against at least one established targeted active learning baseline (such as TALISMAN on KITTI or SIMILAR/ORIENT on classification using target failure exemplars as query sets), or integrate the matched-information controls from Appendix F directly into the primary comparative discussion in Section 5.1.
5. **Lack of Variance Measures and Narrow Margins in Component Ablation (Table 4)**
   - *Description & Context:* Section 5.5 and Table 4 evaluate ImageNet10 classification test accuracy across all scoring component combinations ($none, R, P, C, P + C, R + C, R + P, R + P + C$) across five ratios, concluding that each term is complementary based on average leave-one-out drops of ≈ 0.0094.
   - *Potential Concern:* Unlike Tables 9–12 in Appendix E, Table 4 reports point estimates without standard deviations over the five random seeds. At specific budget ratios, the margin between two-component ablations and the full three-component score is very small. For example, at ratio 0.1, $P + C$ achieves 0.7359 while $R + P + C$ achieves 0.7369—a difference of 0.0010, which corresponds to 1.3 test images on the 1,300-image ImageNet10 test set. Given that seed-to-seed standard deviations in Table 9 range between ± 0.0010 and ± 0.0044, it is unclear whether these narrow differences are statistically distinguishable from run-to-run training noise. Additionally, the fallback selection rule for the "none" baseline (when R, P, and C are all zero, yielding $\tilde{s}_i \equiv 0$) is unspecified.
   - *Suggested Improvement:* Add standard deviations across the five seeds to Table 4, qualify claims regarding component complementarity at small ratios where margins are within empirical variance, and specify the tie-breaking/selection mechanism used for the "none" column.
6. **Synthetic-to-Real Data Ratio Confounder in Small-Dataset Detection (KITTI Ratio 0.1)**
   - *Description & Context:* In Section 5.1 (lines 357–364), all methods yield negative RER on KITTI at ratio 0.1 (-9.65% to -5.81%). The text attributes this degradation exclusively to a "low-data failure mode" where 500 real source images provide insufficient variation.
   - *Potential Concern:* The generated repair pool $|D_{\text{sel}}^{\text{gen}}|$ is held constant at 1,000 images across all tasks and selection ratios (lines 141–142). On KITTI at ratio 0.1 ($n_{\text{select}} = 500$), the retraining set consists of 500 real source images and 1,000 synthetic images, creating an inverted 2:1 synthetic-to-real ratio (66.7% synthetic). In dense object detection, diffusion-generated images can introduce scale distortions, box boundary artifacts, and texture shifts. Retraining YOLOv8n on a dataset dominated 2:1 by synthetic data is a plausible compounding factor for detector degradation, independent of real source sample size.
   - *Suggested Improvement:* Discuss the synthetic-to-real data mixture ratio as a potential compounding factor in the KITTI ratio-0.1 degradation, and evaluate whether scaling $|D_{\text{sel}}^{\text{gen}}|$ proportionally with source sample size mitigates performance regression.

Minor Corrections and Typos

1. **Figure 2 Caption and Seed Clarification:** In Figure 2, the bottom row reports "overall" accuracy differences of 4.1% for Ours, 3.7% for BADGE, and 2.8% for Random. These differ from the five-seed mean accuracy differences computable from Table 9 (where Ours is $0.7508 - 0.7173 = 3.35$, BADGE is 2.96, and Random is 2.07) because Figure 2 visualizes seed 42 (per line 730). Explicitly stating in the caption of Figure 2 that values reflect seed 42 will resolve this discrepancy.
2. **Line 323 / Baseline Metrics Context:** Line 323 states: *"In the main RER tables, M_{before} is 0.7173 for ImageNet10, 0.9057 for CelebA, 0.5902 for COCO, and 0.5626 for KITTI."* Explicitly specifying that these represent *overall test set* performance of the pre-repair checkpoint would prevent ambiguity with slice-specific baselines.
3. **Line 348 Precision:** The text notes that *"PPAL is slightly higher at ratio 0.7 by 0.05 RER points"*. Stating "0.05 RER percentage points" is more mathematically precise. Furthermore, in Table 12, the underlying mAP50–95 metrics are 0.5782 ± 0.0030 for PPAL and 0.5780 ± 0.0039 for Ours, indicating that this difference is within one standard deviation.
4. **Table Reference in Section 5.3 (Line 421):** Line 421 states: *"Because these heatmaps show marginal attribute values, Appendix F also reports the full diagnosed conjunction and its strict complement."* Directly referencing Table 17 rather than the general "Appendix F" would improve document cross-referencing.
5. **Table 2 Unit Formatting:** In Table 2, units are mixed across columns: Entropy, UHerding, and PPAL are listed in seconds (`s`), CoreSet and BADGE switch between minutes (`m`) and hours (`h`), and Ours is in seconds (`s`). Standardizing to seconds or minutes across all entries would improve visual consistency.
6. **Table 3 Column Header Footnote:** In Table 3, the baseline column is labeled `Best AL` . Indicating which active learning baseline achieved the best result at each ratio (e.g., in a table footnote or caption) would make the table fully self-contained.
7. **Line 462 (Numerical Precision):** In Section 5.5, single-component averages are reported to five decimal places ("R: 0.74198, P: 0.74038, C: 0.73814"). Since the ImageNet10 test set contains 1,300 images (where a single sample represents ≈ 0.000769 or 0.077%), reporting five decimal places exceeds the empirical resolution of the test split; rounding to four decimal places would align with Tables 9 and 10.
8. **Figure 2 Colorbar Range:** The colorbar in Figure 2 is labeled "Accuracy Differences (%)" with tick marks from -20 to $+20$, yet all plotted cells are strictly non-negative (0.0 to 17.6). Adjusting the visual colorbar range to $[0, 20]$ would improve visual contrast.

[Extended Empirical Controls, Robustness, and Audits] (Pages: 13-26)

Summary

This segment comprises Appendices E through K (pages 13–26, supported by implementation details in Appendices A–D), presenting extended empirical controls, baseline comparisons, robustness evaluations, and auditing protocols for AttriSelect:

- **Raw Benchmark Metrics and Significance (Appendix E, Tables 9–12):** Details native task metrics (classification test accuracy on ImageNet10 and CelebA, COCO keypoint AP, and KITTI test mAP50–95) across five random seeds (0, 42, 1234, 3407, 2024), supported by paired t-tests ($p < 0.05$) to verify statistical significance across source ratios $r \in 0.1, 0.3, 0.5, 0.7, 0.9, 1.0$.

- **Controlled and Matched-Information Baselines (Appendix F, Tables 13–19):** Compares against attribute-informed controls (`Target−Slice + Stratified` , `Attribute−Filtered Random` , `Attribute−Filtered Entropy` , and `Generated−Complement`), evaluates performance on exact diagnosed failure conjunctions versus non-target data (Table 17), characterizes selected-subset diversity and class balance (Table 18), and examines highly constrained source budgets down to ratios of 0.02 and 0.05 (Table 19).
- **Slice Heatmaps and VLM Auditing (Appendices G & H, Figure 4, Tables 20–21):** Reports a 38-attribute slice AP delta heatmap for COCO keypoint estimation, outlines data leakage prevention protocols, details a blinded five-annotator human audit across 100 images per benchmark (Table 20), and examines sensitivity to synthetic attribute label corruption at 5%, 10%, and 20% noise levels (Table 21).
- **Pipeline and Generator Transferability (Appendix I, Tables 22–24):** Tests selector stability across upstream repair pipelines (HiBug on CelebA) and generative backbones (InstructPix2Pix on CelebA, ICEdit on COCO).
- **Sensitivity Studies and Standalone Active Learning (Appendices J & K, Tables 25–32):** Assesses sensitivity to retraining duration (15 to 60 epochs), slice-granularity definitions (single-field vs. multi-field conjunctions), and missing attribute fields. Evaluates AttriSelect in source-only regimes without repair checkpoints or generative data in one-shot subset selection and iterative pool-based active learning across multiple VLM taggers (Qwen3-VL, BLIP, InternVL3).

Potential Mistakes and Improvements

1. **Underspecified Candidate Class Conditioning and Potential Label Leakage in Iterative Active Learning (Appendix K.2)**
   - *Context:* Appendix K.2 (lines 1383–1386) evaluates AttriSelect (`AttriSelect−AL`) across five active acquisition rounds where *"only the class label is queried"* from an unlabeled pool.
   - *Potential Concern:* In Equation (2) and Algorithm 1, AttriSelect's selection scoring relies on candidate x_i's class label y_i to evaluate class-conditional marginal rarity $R_{i,a} = (n_a(v_{i,a} \mid y_i) + \varepsilon)^{-\alpha}$ and to assign candidate x_i to class $c = y_i$ for the soft class-balancing correction $C_c^{(k)}$. In a standard pool-based active learning setup, the ground-truth class labels for unselected candidates in the unlabeled pool are unobserved prior to query acquisition. The text does not specify whether candidate scoring uses model-predicted pseudo-labels $\hat{y}*i* = *arg\max f*\theta(x_i)$, marginalizes over predicted class posteriors, or falls back to unconditional marginal counts. If ground-truth labels were accessed during candidate scoring this would constitute label leakage relative to the baseline active learning methods. If pseudo-labels were used, early-round classification errors (where initial ImageNet10 accuracy is 43.23%) would introduce noise into both rarity estimation and class balancing.
   - *Suggested Improvement:* Document the exact scoring formulation used for unselected candidate samples in Appendix K.2, detailing how candidate class conditioning is handled prior to que acquisition.
2. **Mathematical Contradiction in Source-Only Scoring Formulation (Appendix K.1 vs. Appendix L.1 and Table 30)**
   - *Context:* In Appendix K.1 (lines 1348–1349), the text states: *"In this setting, the generated-data frequency term is omitted and selection uses the available source attributes and soft class balancing, as formalized in Appendix L."* In lines 1367–1368, a separate baseline named `Attribute rarity` is defined as ranking images *"by class-conditional rarity of their assigned attribute values and applies soft class balancing."*
   - *Potential Concern:* If the frequency penalty were omitted from AttriSelect ("Ours"), the scoring function would be mathematically identical to the `Attribute rarity` baseline. In Table 30, however, `Attribute rarity` and `Ours` report distinct test accuracies across all five ratios (e.g., at ratio 0.1: 44.46% vs. 45.00%; at ratio 0.5: 65.31% vs. 66.53%). Appendix L.1 (Equation 27, lines 1640–1646) clarifies that when $\mathcal{D}*{\text{gen}}* = *emptyset*, $the second term is not omitted, but ratheracts as a source-distribution head-value suppression penalty$: $*bar{s}i* = *sum{a \in *mathcal{A}} *left[*left(*frac{1}{n_a(v*{i,a} \mid y_i) + *varepsilon}\right)^\alpha - \lambda \frac{n_a^{\text{src}}(v_{i,a})}{\sum_{v \in V_a} n_a^{\text{src}}(v)} \right]$$
   - *Suggested Improvement:* Reconcile the narrative in Section K.1 (lines 1348–1349) with Equation (27) by clarifying that `Ours` retains the source-pool head-value suppression penalty with weight $\lambda > 0$, whereas the `Attribute rarity` baseline corresponds to the ablation where $\lambda = 0$.
3. **Potential Metric Inconsistency for the COCO Exact Target Slice in Table 16**
   - *Context:* Table 16 (line 1031) summarizes the `Generated−Complement` control, stating in the caption that *"Exact-target, non-target, and overall entries use each task's native metric."* For COCO, the table lists the `Generated−Complement` exact-target metric as **0.7261**.
   - *Potential Concern:* In Table 17 at ratio 0.5, the COCO exact target conjunction ("brown clothes AND rainy weather", $n = 22$) keypoint AP is reported as **0.4813** pre-repair, **0.5680** for `Generated−Complement` , and **0.5826** for AttriSelect. Across all main COCO evaluations (Tables 1 and 11), overall keypoint AP across all ratios and baselines ranges between 0.5877 and 0.635 A five-ratio mean keypoint AP of 0.7261 on this diagnosed underperforming failure slice appears inconsistent with the ratio-0.5 exact slice score and the overall AP scale, suggesting an unnoted metric substitution (such as $\text{AP}*{50}* instead of *text{mAP}*{50\text{--}95}$) or a reporting typo.
   - *Suggested Improvement:* Verify the metric computation for the COCO entry in Table 16 and align the reported value or metric definition with Table 17.
4. **Protocol Ambiguity and Omitted Target Field in Human VLM Audit (Appendix H, Table 20)**
   - *Context:* Appendix H reports a blinded human audit involving five annotators across 100 images per dataset (Table 20).
   - *Potential Concern:*
     1. In line 1110, the CelebA repair slice is defined as a three-field conjunction: *red hair, brown skin, and sad emotion*. Table 20 reports agreement for `Hair color: 98%` and `Skin tone: 99%` but omits the agreement percentage for the third field (`emotion`).
     2. Because targeted failure slices represent rare subpopulations (empirical base rates around 1%–2%), drawing 100 images uniformly at random yields only 1 to 2 positive slice instances i expectation. In that setting, high binary slice-indicator agreement (90%–94%) can be heavily dominated by true negatives.
     3. ImageNet10 is the primary testbed for core classification repair and ablations, yet it is excluded from both the human audit (Table 20) and the noise sensitivity sweep (Table 21), despite relying entirely on VLM-assigned metadata.
   - *Suggested Improvement:* Report the human agreement percentage for facial emotion on CelebA in Table 20; clarify whether the 100 audit samples were drawn uniformly or enriched for candidate target slices; report chance-adjusted agreement metrics (such as Cohen's Kappa) or precision/recall; and extend the audit and noise sensitivity sweep to ImageNet10.
5. **Degradation Under 20% Attribute Noise and Rarity Weighting of Outliers (Appendix H, Table 21 & Appendix P)**
   - *Context:* Table 21 evaluates sensitivity to independent attribute label corruption at 5%, 10%, and 20% noise levels at source ratio 0.5.
   - *Potential Concern:* Introducing 20% noise causes an absolute performance drop of 0.090 (9.0 percentage points) on CelebA accuracy and 0.060 (6.0 mAP points) on KITTI. On CelebA, this dro reduces clean accuracy from 0.9187 to ~0.8287, falling below the un-repaired checkpoint (0.9057) and below random selection (0.9114). On KITTI, mAP drops from 0.5773 to 0.5173, below tl initial checkpoint (0.5626). Because the class-conditional marginal rarity term $R_{i,a} = (n_a(v_{i,a} \mid y_i) + \varepsilon)^{-\alpha}$ assigns the highest priority to low-frequency values, corrupted or misassigned attribute tokens naturally generate artificial low-frequency singletons that are actively prioritized by the selector.
   - *Suggested Improvement:* Discuss this noise-magnification characteristic in Appendix P (Limitations), and discuss potential mitigations, such as frequency thresholding or continuous confidence weighting in $R_{i,a}$.
6. **Absence of Dedicated Targeted Active Learning and Submodular Baselines (Appendix F.1)**
   - *Context:* In Section 2 (lines 95–99), the manuscript cites frameworks specifically developed for targeted active learning and distribution-shift subset selection, such as TALISMAN [Kothawade et al., ECCV 2022] for rare slices in detection and ORIENT [Karanam et al., NeurIPS 2022] for subset selection under distribution shift via Submodular Mutual Information (SMI).
   - *Potential Concern:* In Appendix F (Tables 13–15), AttriSelect is compared against generic active learning methods (CoreSet, BADGE, PPAL, UHerding) and heuristic controls (`Target+Stratified` , `Filtered Random` , `Filtered Entropy` , `Generated−Complement`). Dedicated targeted active learning frameworks optimizing submodular mutual information conditioned on target exemplars are not included.
   - *Suggested Improvement:* Benchmark against at least one established targeted subset selection baseline (e.g., TALISMAN on KITTI or Facility Location SMI / ORIENT on ImageNet10), or provic a formal rationale detailing the operational constraints that distinguish AttriSelect from submodular mutual information approaches.

Minor Corrections and Typos

1. **Algorithm 1 Candidate Exclusion Set Notation (Page 16, Lines 844 & 848):** In steps 11 and 15 of Algorithm 1, the candidate exclusion set is typeset with a forward slash as `$x_i \in D/_{\text{sel}}^{\text{src}}$` and `$\arg\max_{x_i \in D/_{\text{sel}}^{\text{src}}}$` . Standard set difference notation should be used instead: $x_i \in D_{\text{src}} \setminus D_{\text{sel}}^{\text{src}}$.
2. **Table 15 vs. Text Discrepancy (Page 19, Line 1009):** The text states: *"Generated-Complement is higher only on KITTI at ratio 0.7, by 0.54 RER points."* In Table 15 at ratio 0.7, AttriSelect achieves an RER of 3.52%, which is also exceeded by PPAL (3.57%) and Target+Stratified (3.54%). The sentence should be rephrased to avoid implying that AttriSelect was the second-highest method at that ratio.
3. **Draft Editorial Note Artifact (Page 24, Line 1250):** The standalone text immediately preceding Section I.2 (*"HiBug table: changing the upstream repair pipeline changes the generated set and can shift every selector's absolute RER."*) appears to be an unformatted draft note and should be integrated into the surrounding narrative of Section I.1.

4. **Undefined Metric: "Class Imbalance" (Table 18, Page 20):** Table 18 reports integer values for "Class imbalance" (ranging from 0 to 6), but the exact mathematical metric (e.g., $\max_c n_c - \min_c n_c$ or $\max_c |n_c - t|$) is not defined in Section F.4 or the caption.
5. **Textual Discrepancy in Table 23 Description (Page 24, Lines 1262–1264):** The text states: *"Its absolute RER is lower than the corresponding ControlNet result in Table 1, and the same decrease appears across the other selectors."* In Table 23 at ratio 0.7, `Ours (InstructPix2Pix)` achieves an RER of **12.13%**, which is slightly higher than `Ours (ControlNet)` (**12.09%** in Table 1). The text should note this minor exception.
6. **Table 19 Column Formatting (Page 21, Line 1088):** The column header `Filt. Rand./Ent.` combines two baselines into single slash-separated cells (`0.68±0.94/–0.95±1.07`). Clarifying the notation in the caption or splitting it into two dedicated columns would improve readability.

[Supplementary Analyses, Visualizations, and Critical Limitations] (Pages: 25-26, 32-36)

Summary

This segment comprises the concluding supplementary analyses, robustness benchmarks, hyperparameter specifications, qualitative embeddings, compliance disclosures, and practical limitations the AttriSelect framework:

- **Slice Granularity and Incomplete Vocabulary Sensitivity (Appendices J.2–J.3, Pages 25–26):** Evaluates performance under varying slice definitions (single-attribute vs. two-attribute vs. three-attribute conjunctions in Tables 26–27) and assesses degradation when key attribute fields are omitted during selection (Tables 28–29).
- **Source-Only Subset Selection and Active Learning (Appendix K, Pages 25–26):** Evaluates the selection mechanism in isolation from generative repair on ImageNet10 across one-shot data selection (Table 30), iterative active learning (Table 31), and sensitivity across vision-language annotators (Qwen3-VL, BLIP, InternVL3 in Table 32).
- **Hyperparameters, Sensitivity, and Component Ablations (Appendices M–N, Pages 32–33):** Documents default parameters (α, λ, w) across benchmarks (Table 33), sensitivity sweeps for KITT and COCO (Tables 34–35), and a complete leave-one-out and single-term score ablation for COCO person keypoint estimation (Table 36).
- **Qualitative Embeddings and Disclosures (Page 10, Appendices O–P, Pages 33–36):** Visualizes proxy t-SNE embeddings of selected subsets against active learning baselines on COCO and KITT (Figures 5–8), provides ethics, reproducibility, and AI-use statements (Page 10), and details practical limitations regarding VLM dependencies, closed-set vocabularies, marginal versus conjunctio rarity, and generative synthesis artifacts (Appendix P).

Potential Mistakes and Improvements

1. **Contradiction in Objective Formulation for Source-Only Selection (Section K vs. Appendix L.1 and Table 30)**
   - *Observation:* In Section K (lines 1348–1349), the text states: *"In this setting, the generated-data frequency term is omitted and selection uses the available source attributes and soft class balancing, as formalized in Appendix L."* However, in Appendix L.1 (Eq. 27, page 31), the formalization of the no-generation corner case ($\mathcal{D}_{\text{gen}} = \emptyset$) $does not omit the frequency penalty; instead, it sets \hat{n}_a(v) = n_a^{\text{src}}(v), converting the penalty into a source-distribution head-value suppression term: $\bar{s}_i = \sum_{a \in \mathcal{A}} \left[\left(\frac{1}{n_a(v_{i,a}) \mid y_i) + \varepsilon}\right)^\alpha - \lambda \frac{n_a^{\text{src}}(v_{i,a})}{\sum_{v \in \mathcal{V}_a} n_a^{\text{src}}(v)} \right]$$ Furthermore, in Table 30 (page 26), the authors report results for both an "Attribute rarity" baseline (defined on lines 1367–1368 as ranking by class-conditional rarity plus soft class balancing, i.e., $R + C$) and "Ours". Across all ratios in Table 30, "Ours" achieves higher accuracy than "Attribute rarity" (e.g., 66.53% vs. 65.31% at ratio 0.5).
   - *Potential Concern:* If the frequency penalty were omitted in Section K as stated on page 25, "Ours" would be mathematically equivalent to the "Attribute rarity" baseline. The distinct scores ir Table 30 suggest that "Ours" in fact incorporates the source head-value penalty from Eq. 27.
   - *Suggested Improvement:* Clarify the exact scoring function and hyperparameter values (α, λ, w) used for "Ours" in Table 30, and reconcile the text in Section K with the mathematical specification in Appendix L.1.
2. **Attribute Vocabulary Mismatch in ImageNet10 Granularity Analysis (Table 27, Page 25)**
   - *Observation:* In Table 27, the slice-granularity ablation for ImageNet10 reports accuracy across four conditions: "Pink only", "Fabric only", "Pink+fabric+rainy", and "Pink+fabric". However, Section 4.1 (line 151), Appendix D (lines 860–863), and Figure 2 define the ImageNet10 attribute fields exclusively as `color` (tan, green, pink, orange, yellow, purple, black), `background` (sk house, buildings, rocks, street), and `texture` (leather, metallic, wooden, fabric). The attribute `rainy` belongs to the weather field used in COCO (Figure 4) and KITTI (Figure 3).
   - *Potential Concern:* Including "Pink+fabric+rainy" on ImageNet10 appears to be an uncorrected artifact from the COCO granularity experiment (Table 26, which evaluates "Brown+rain+night" A reader cannot determine which background or contextual attribute was actually queried and evaluated for the three-field condition on ImageNet10.
   - *Suggested Improvement:* Correct the column header and text to reflect the actual third attribute field evaluated (e.g., `Pink+fabric+street` or `Pink+fabric+rocks`).
3. **Unpopulated Metric Placeholders (`nan`) in KITTI Small-Budget Visualizations (Figure 7, Page 35)**
   - *Observation:* In Figure 7, each of the eight subpanels visualizing KITTI ratio-0.05 selections displays an unpopulated metric in its title string: `Random n=250 mAP50–95=nan` , `Stratified n=25 mAP50–95=nan` , `Entropy n=250 mAP50–95=nan` , `CoreSet n=250 mAP50–95=nan` , `BADGE n=250 mAP50–95=nan` , `PPAL n=250 mAP50–95=nan` , `UHerding n=250 mAP50–95=nan` , and `Ours n=250 mAP50–95=nan` . In contrast, Figure 6 (ratio 0.10) reports evaluated numerical mAP values matching Table 12.
   - *Potential Concern:* Table 19 evaluates ratio 0.05 on ImageNet10 and COCO but omits KITTI. Displaying `nan` across all headers in Figure 7 indicates an unhandled logging artifact and leaves ambiguity as to whether models were trained and evaluated on these 250-image subsets.
   - *Suggested Improvement:* If evaluation was conducted, report the actual test mAP50–95 values in the headers; otherwise, remove the metric string from the panel titles and clarify in the caption that Figure 7 displays spatial selection geometry without downstream evaluation.
4. **Unexplained Discrepancy in Selected Sample Counts in COCO Visualizations (Figure 5, Page 34)**
   - *Observation:* In Figure 5 (COCO ratio 0.10), the total source pool contains $N = 20,000$ training images (line 310), which corresponds to a nominal budget of $n_{\text{select}} = 2,000$ images. However, the subpanels report divergent sample sizes across methods: `Random n=1970` , `Stratified n=1206` , `Entropy n=1206` , `CoreSet n=1151` , `BADGE n=1185` , `PPAL n=1198` , `UHerding n=1184` , and `Ours n=1253` . In contrast, Figure 6 (KITTI ratio 0.10) lists exactly $n = 500$ across all eight methods.
   - *Potential Concern:* The text in Section O does not state what n represents in Figure 5. If different active learning selectors were allocated unequal numbers of source images during retrainin this would confound the experimental comparison. Alternatively, if n represents the count of person keypoint instances or valid proxy embeddings plotted after filtering, this is not documented.
   - *Suggested Improvement:* Clarify in the caption of Figure 5 or Section O whether n denotes image counts or valid plotted instances, and confirm that all models were trained under identical source-data budgets.
5. **Numerical Inconsistency in COCO Component Ablation (Table 36 vs. Tables 11 and 35)**
   - *Observation:* For COCO pose-estimation repair at ratio 0.7, the performance of the full AttriSelect configuration ($R + P + C$) is reported as:
     - 0.6319 ± 0.0024 in Table 11 (Main COCO repair results, page 18);
     - 0.6319 in Table 35 (COCO AttriSelect sensitivity under default hyperparameters $\alpha = 0.03, \lambda = 0.5, w = 15$, page 33);
     - 0.6311 in Table 36 (COCO component ablation under the $R + P + C$ column, page 33).
   - *Potential Concern:* While the numerical margin (0.0008 AP) is small, this causes the class-balancing term alone ($C = 0.6319$) in Table 36 to appear higher than the full model ($R + P + C = 0.6311$) at ratio 0.7. Reporting different metrics for the identical checkpoint and parameter configuration across adjacent tables creates an internal contradiction.
   - *Suggested Improvement:* Verify the evaluation logs across random seeds at COCO ratio 0.7 to ensure consistent reporting across Tables 11, 35, and 36.
6. **Scope Constraints of Additive Marginal Scoring on Conjunction Slices (Appendix P)**
   - *Observation:* Throughout the paper, failure slices are defined as multi-attribute conjunctions (e.g., ImageNet10: pink color AND fabric texture; COCO: brown clothes AND rainy weather; CelebA: red hair AND brown skin AND sad emotion). However, the selection score sums 1D marginal frequencies: $\bar{s}_i = \sum_{a \in \mathcal{A}} (R_{i,a} + P_{i,a})$. In Appendix P (lines 1900–1906), the authors acknowledge: *"The current score sums per-field marginal counts. A diagnosed conjunction can therefore be rare even when each of its marginal attribute values is common. In that case, the pairwise marginal-rarity condition in Appendix L may fail, and the current theory does not guarantee enrichment of the conjunction."*
   - *Potential Concern:* In practical vision applications, systematic failures frequently arise from conjunctions of individually common attributes (e.g., common vehicle color paired with common daytime lighting). In such cases, marginal frequencies remain high and relatively uniform, rendering the marginal rarity term unable to prioritize the failure conjunction over nominal data.
   - *Suggested Improvement:* Expand the discussion in Appendix P to address the statistical trade-offs between joint conjunction counts ($n_{a_1,\ldots,a_k}(v_1, \ldots, v_k \mid y_i)$) and marginal counts, specifically regarding histogram sparsity and sample complexity under high-dimensional attribute vocabularies.

Minor Corrections and Typos

1. **Unformatted Script Export Strings in Figure Layouts:**
   - In Figures 5, 6, and 7, internal logging lines and data export strings appear verbatim in the document:
     - Figure 5 (line 1801): `original candidates. Colored: selected subset. Sources: Random: selection_exports; Stratified: selection_exports; Entropy: selection_exports; CoreSet: selection_exports; BADGE: selection_exports; PPAL: selection_exports; UHerding: selection_exports; Ours: sele`
     - Figure 6 (line 1829): `subset. Sources: Random: exact selected_original_ids; Stratified: exact selected_original_ids; Entropy: exact selected_original_ids; CoreSet: exact selected_original_ids; BADGE: exact selected_original_ids; PPAL: exact selected_original_ids; UHerding: exact sele`
     - Figure 7 (line 1853): `subset. Sources: Random: exact selected_original_ids; Stratified: exact selected_original_ids; Entropy: exact selected_original_ids; CoreSet: exact selected_original_ids; BADGE: exact selected_original_ids; PPAL: exact selected_original_ids; UHerding: exact sele`
   - *Correction:* Remove these uncurated terminal/script export fragments from the figure environments.
2. **Self-Referential Citation in Proposition 7 Proof (Line 1661, Page 31):**
   - Line 1661 states: *"The proof is identical to the rare-slice enrichment proof in Appendix L..."*
   - *Correction:* Because this statement is already located inside Appendix L (Section L.1), it should explicitly reference Proposition 3 on page 29 rather than referencing "in Appendix L" generally.
3. **Table Float Placement across Page Boundaries (Pages 32–33):**
   - The introductory sentence for Table 35 appears at line 1721 on page 32 (*"Table 35 shows the same pattern on COCO pose-estimation repair..."*), but Table 35 is rendered at the top of page 33, separated from its referencing text.
   - *Correction:* Adjust float parameters so that Table 35 appears adjacent to its introducing paragraph.

LLM Feedback by Program Chairs

LLM Feedback by Program Chairs 15 Sept 2026, 22:47 Conference, Authors

Feedback:
Hello!

```
You requested feedback on your paper submitted to ICLR using the Google Paper Assistant Tool (PAT).
We received your paper and are currently processing it. We will post the feedback here once it is ready.

If you have not received feedback within 24 hours, please contact us at paper-assistant@google.com.
```

About OpenReview (/about) Contact (/contact)
FAQ (https://docs.openreview.net/getting-started/frequently-asked-questions) Hosting a Venue (/group?id=OpenReview.net/Support)
Sponsors (/sponsors) Terms of Use (/legal/terms) / Privacy Policy (/legal/privacy)
All Venues (/venues) **Donate** (/donate)

OpenReview (/about) is a long-term project to advance science through improved peer review with legal nonprofit status. We gratefully acknowledge the support of the OpenReview Sponsors (/sponsors). © 2026 OpenReview